

# JAYANTH KUMAR PILLAJETTI

AI Engineer · AI automation · Agentic AI · Conversational AI  
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## PROFESSIONAL SUMMARY

AI Engineer specializing in intelligent automation with 2+ years of experience in Software engineering and building LLM-based agents, RAG architectures, and complex workflow automation using Python and n8n. Expert in LangGraph, multi-agent orchestration, and prompt engineering, with a proven track record of reducing manual operational effort by 70–80%. Published AI patent holder adept at building reliable data extraction pipelines and seamlessly integrating AI into enterprise systems.

## TECHNICAL SKILLS

**Languages & Core Tech:** Python, SQL, PL/SQL, RESTful API

**AI & Machine Learning :** LLMs, SLMs, Prompt Engineering, Context Engineering, RAG Architectures, Vector Databases, Embeddings, Natural Language Processing (NLP), Machine Learning

**Automation & Agentic Frameworks:** n8n, AI Agents, Multi-Agent Workflows, LangChain, LangGraph

**Deployment:** Docker, AWS, CI/CD Fundamentals, Git

**Certificates:** Prompt Engineering for developers - Deeplearning.ai

## WORK EXPERIENCE

**Tata Consultancy Services (TCS)** | *AI Engineer*

Apr 2024 – Present

- Architected and deployed an enterprise LLM-powered AI assistant leveraging Retrieval-Augmented Generation (RAG), semantic search, vector embeddings, prompt engineering, and contextual reasoning to automate enterprise knowledge retrieval, enabling intelligent decision support and reducing manual operational effort.
- Engineered production-grade AI pipelines for document ingestion, embedding generation, vector indexing, retrieval optimization, context orchestration, and grounded LLM inference, improving response accuracy, minimizing hallucinations, and delivering scalable AI-powered automation.
- Optimized AI system performance through prompt engineering, retrieval tuning, knowledge grounding, LLM evaluation, response quality optimization, and continuous feedback loops, enabling reliable automation of high-volume enterprise support workflows and improving first-response efficiency.

## PROJECTS

**Autonomous AI Job Hunter Agent** | *Python · LangGraph · LangChain · RAG · FAISS · n8n · Gemini API*

- Engineered an autonomous system utilizing LangGraph, FastAPI, and n8n to fully automate ATS resume optimization and job tracking. Developed a gemini-2.5-flash agent that dynamically fetches real-world application updates via Google Sheets and Gmail APIs.
- Designed a real-time data ingestion pipeline using the Apify LinkedIn Jobs Scraper. Implemented a local qwen2.5:3b LLM classifier to filter irrelevant roles, strategically reducing API compute costs before routing data to heavier Gemini models.
- Orchestrated a resume generation engine using robust prompt engineering to strictly enforce structured JSON outputs. Automated targeted ATS keyword injection, Gutenberg HTML-to-PDF conversion, and direct Google Drive synchronization without hallucinating facts.

**Enterprise AIOps Co-Pilot** (Pulseops AI) | *Python · FastAPI · Multi-Agent AI · RAG · ChromaDB · Llama*

- Built an enterprise-grade Agentic AI platform orchestrating specialized LLM agents for email understanding, incident prioritization, knowledge retrieval, and intelligent engineer allocation across IT operations.
- Designed a self-learning RAG system using vector embeddings, semantic retrieval, metadata-aware search, and continuous knowledge indexing to provide accurate root-cause analysis and remediation recommendations.
- Engineered a scalable AI backend integrating FastAPI, vector databases, and real-time dashboards to automate incident management workflows while improving SLA compliance and reducing mean time to resolution (MTTR).

## EDUCATION

**SRM University Andhra Pradesh**

Aug 2019 – May 2023

*B.Tech, Computer Science — AI & Machine Learning Specialisation*

*GPA: 8.2 / 10*

## ACHIEVEMENTS

- Published Patent:** "System and Method for Heart Disease Prediction Using Supervised Machine Learning Algorithms" — Indian Patent Journal; engineered a unified dataset by merging multiple clinical data sources, improving data quality and achieving high model accuracy for reliable early-stage prediction.
- Spot Award (TCS):** Awarded for exceptional performance in developing AI-driven solutions, contributing to automation and improved operational efficiency.